

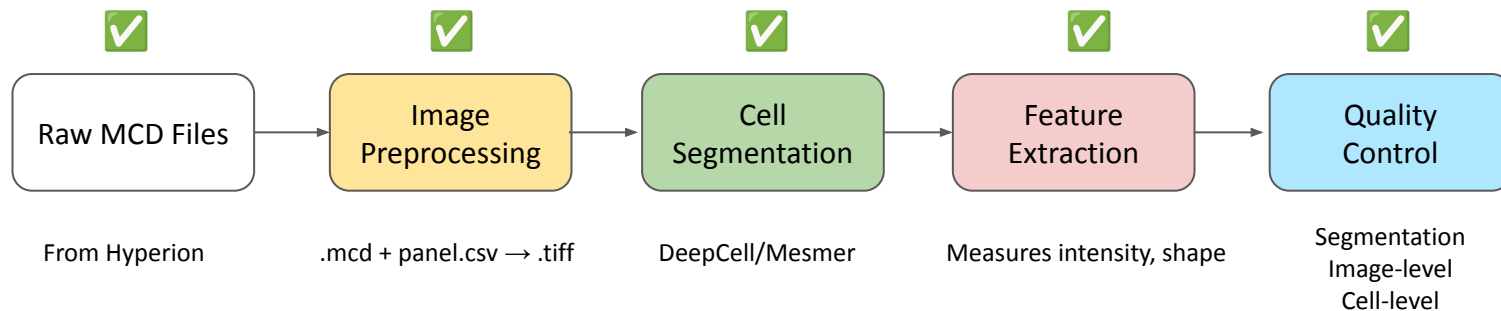
6/12

NSCLC Tumor IMC Analysis Updates

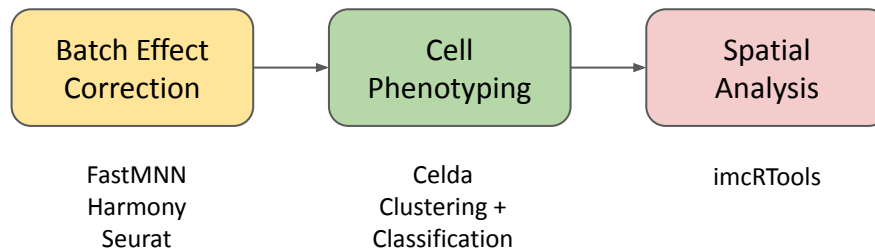
Tazein Shah, Vaidehi Gupta
BMSIP 2026

IMC Data Analysis - Progress

Completed Steps (Patient 12 + 14)



Next Steps (Post-steinbock on all patients + QC)

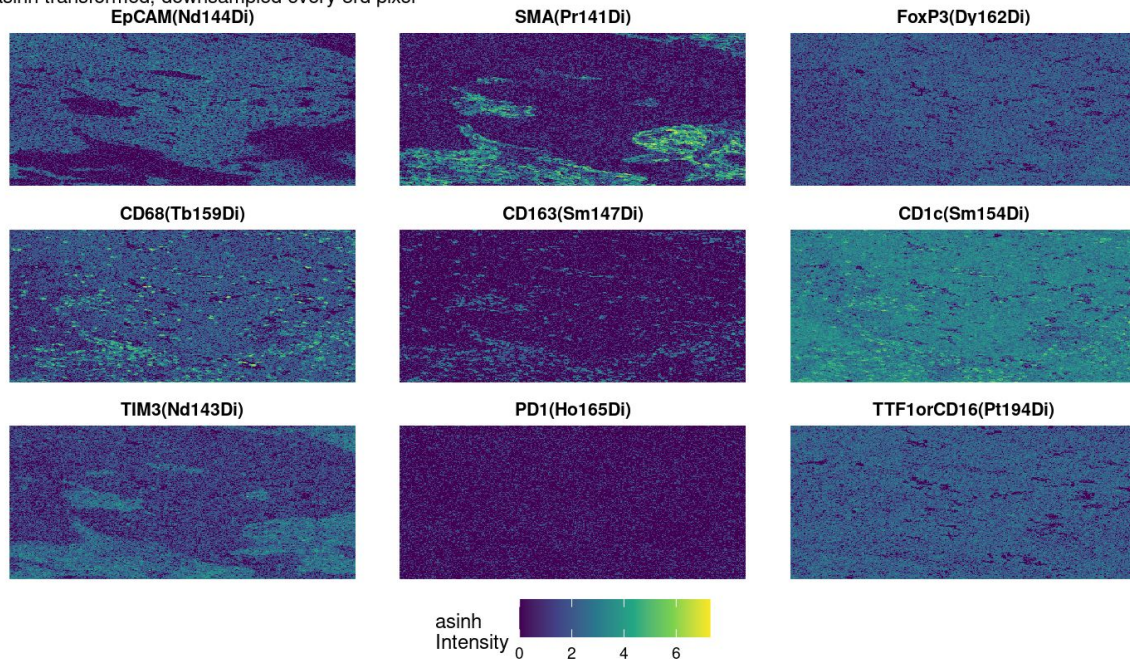


Raw Data Spatial Plots

- Lot of background noise

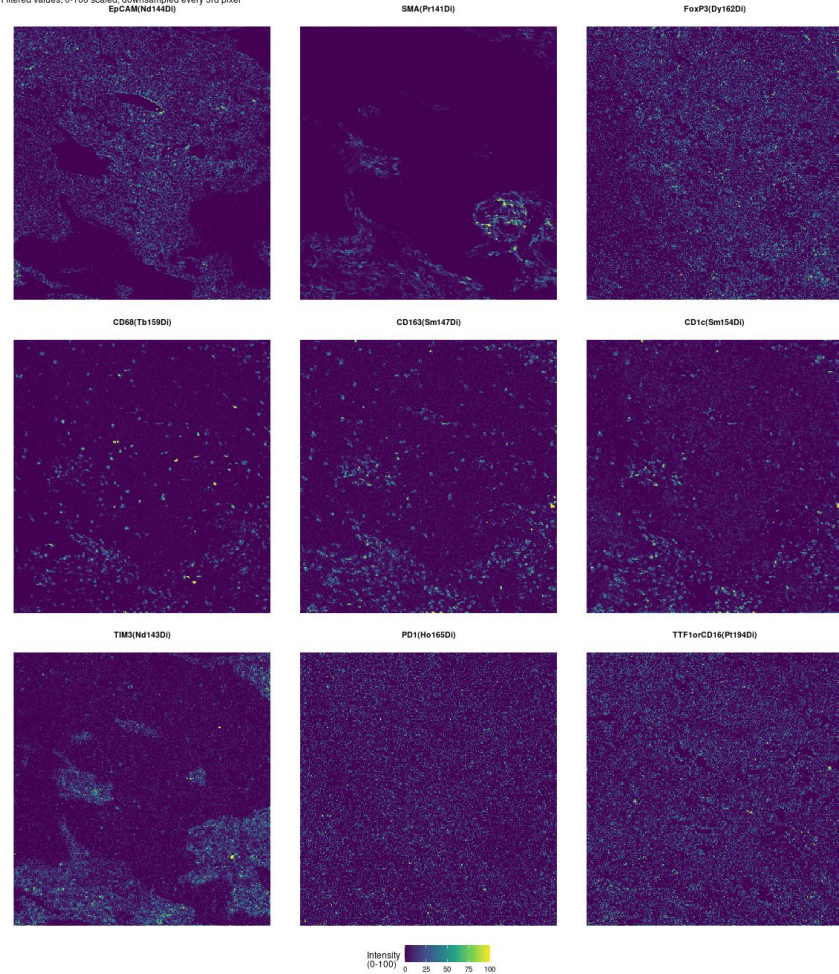
Patient 12 ROI 1 - Selected Immune Markers

asinh transformed, downsampled every 3rd pixel



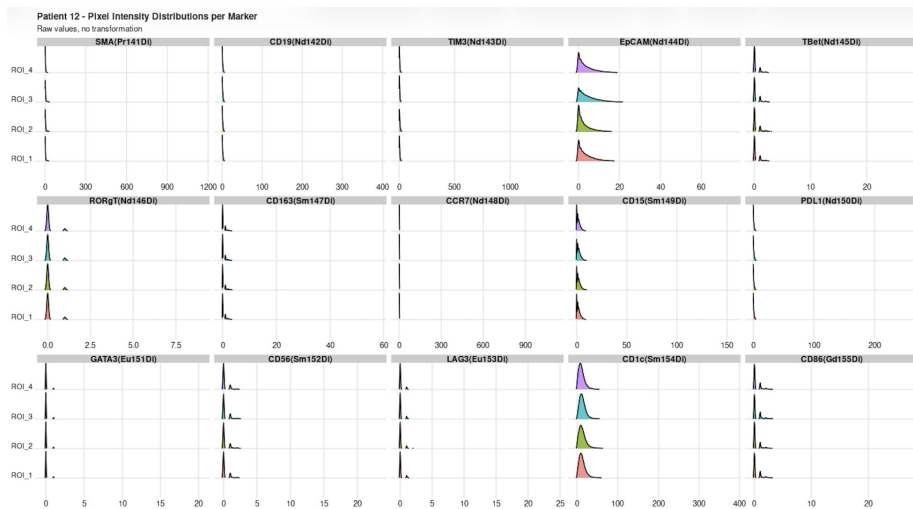
Filtered Data Spatial Plots

- GATA3 added to panel.csv
- Less background noise compared to raw data
- Can see discrete bright spots across the tissue

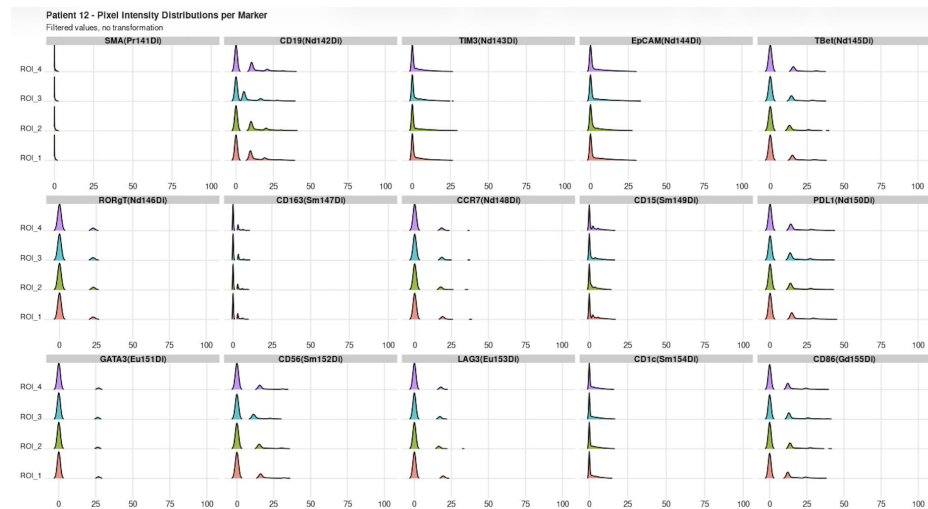


Pixel Intensity Distributions - Raw vs. Filtered Data

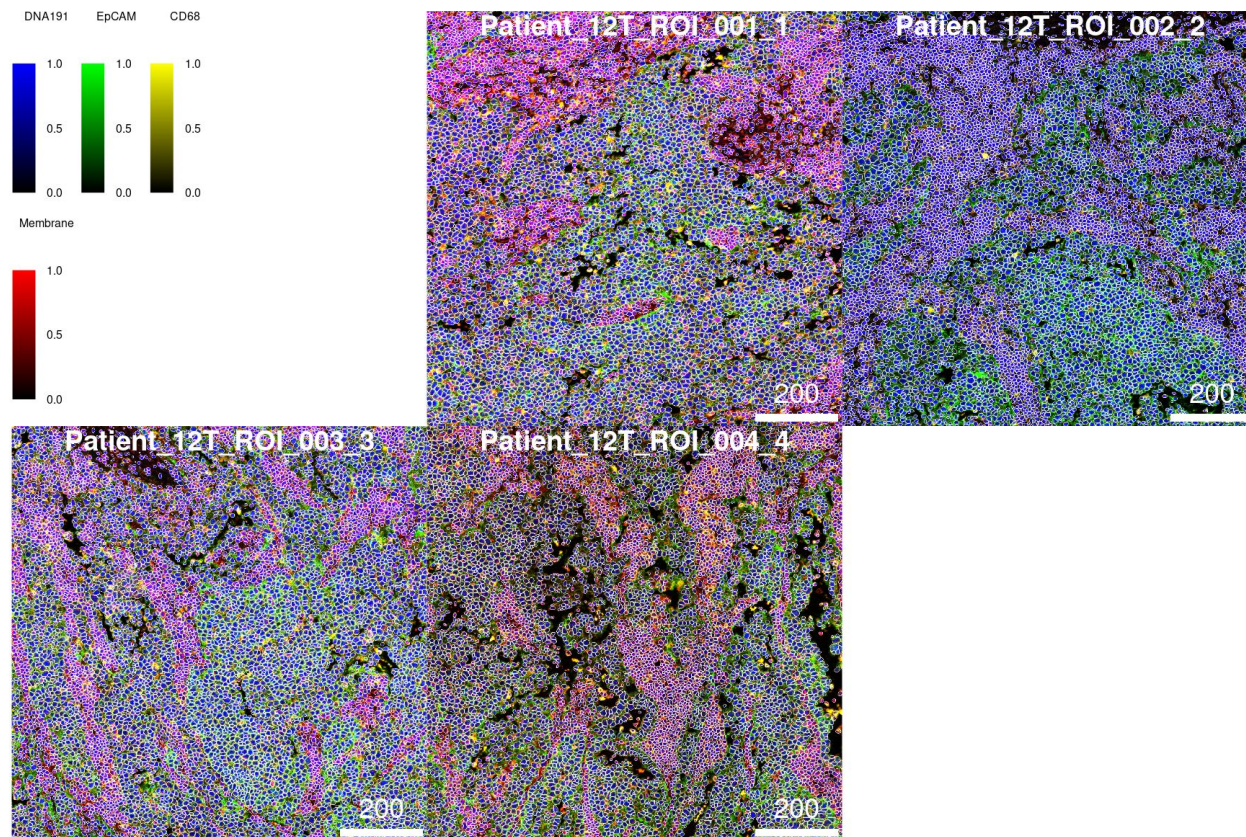
Raw data



Filtered data



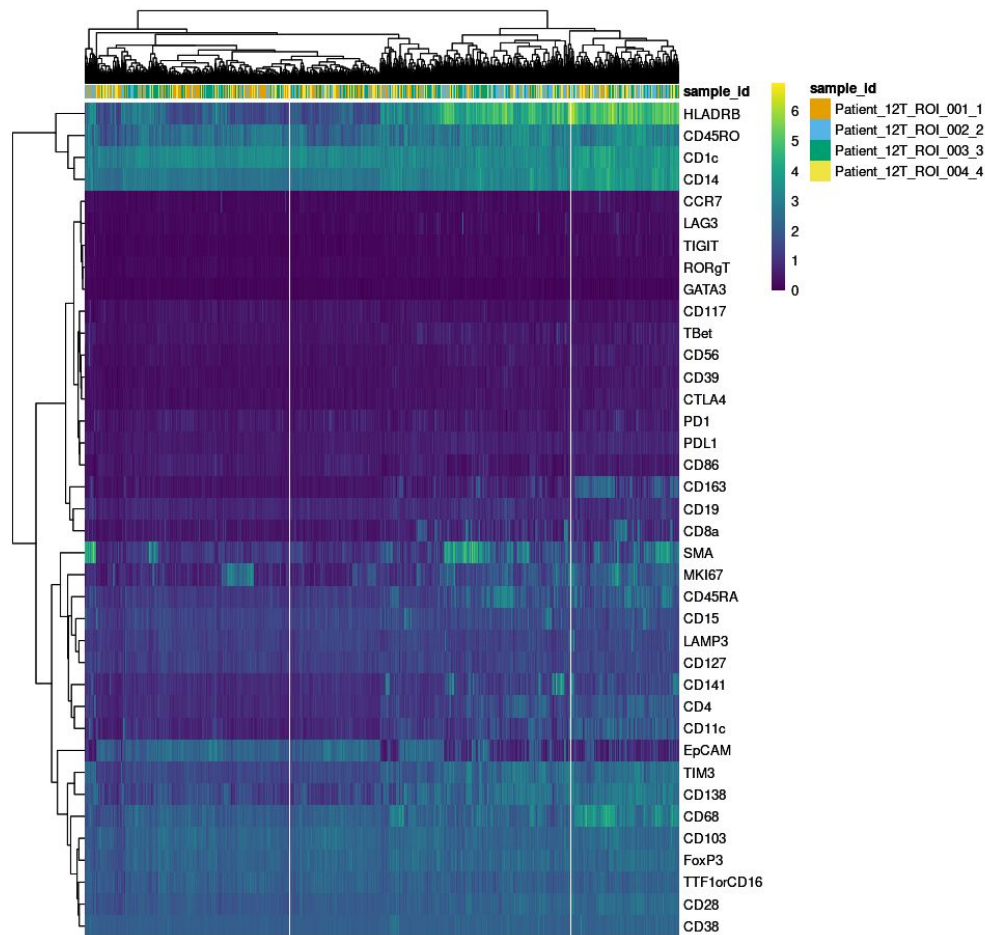
Cell Segmentation - Patient 12 LCNEC (High Malignancy) + Adenocarcinoma



Patient 12: Single-cell Marker Expression by ROI

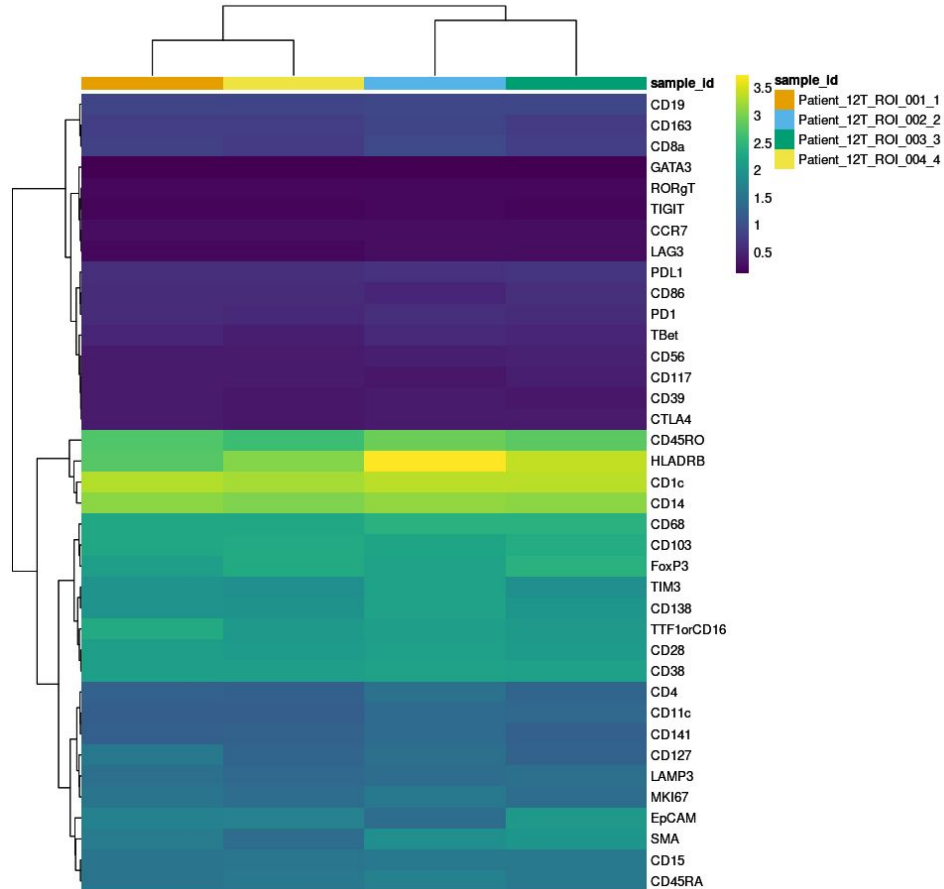
Subsampled to 2000 cells

asinh transformed

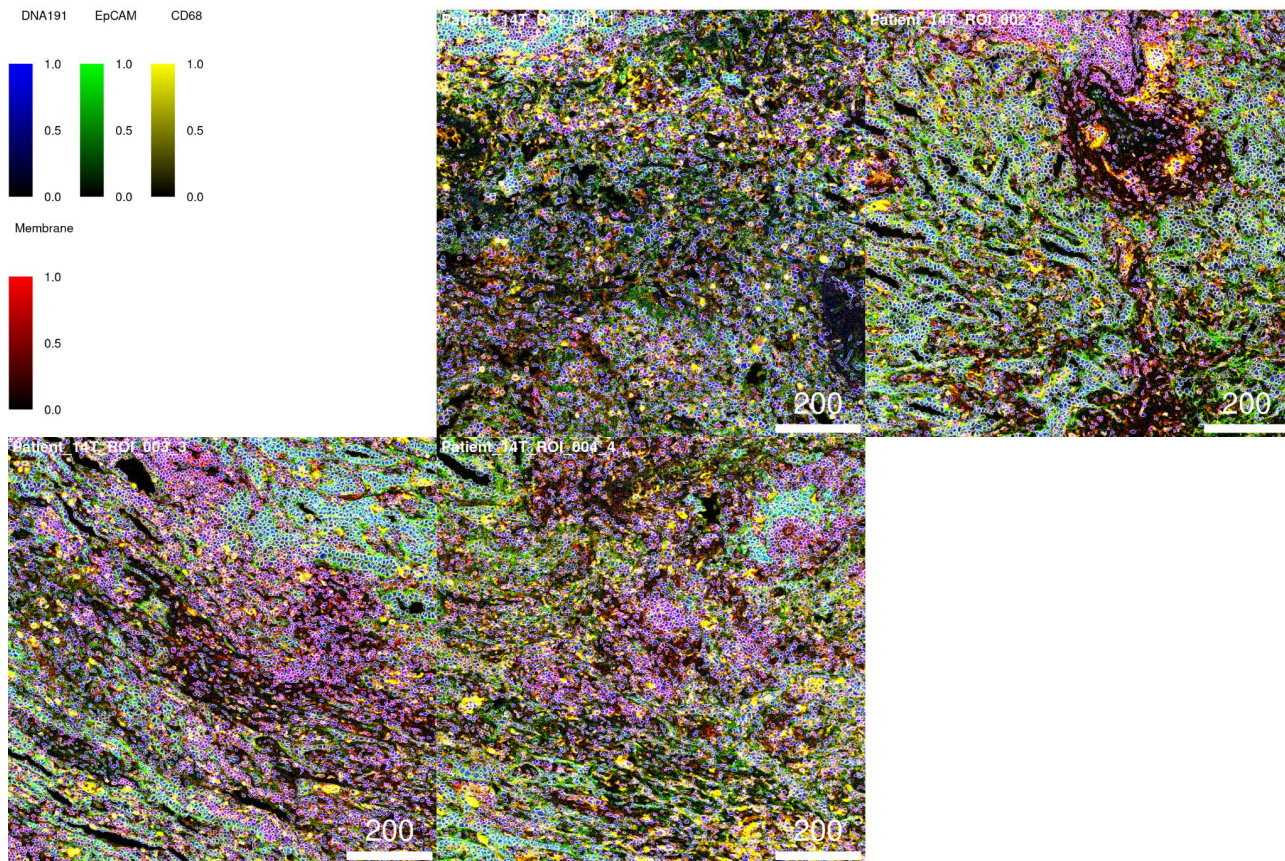


Patient 12: Average Marker Expression per ROI

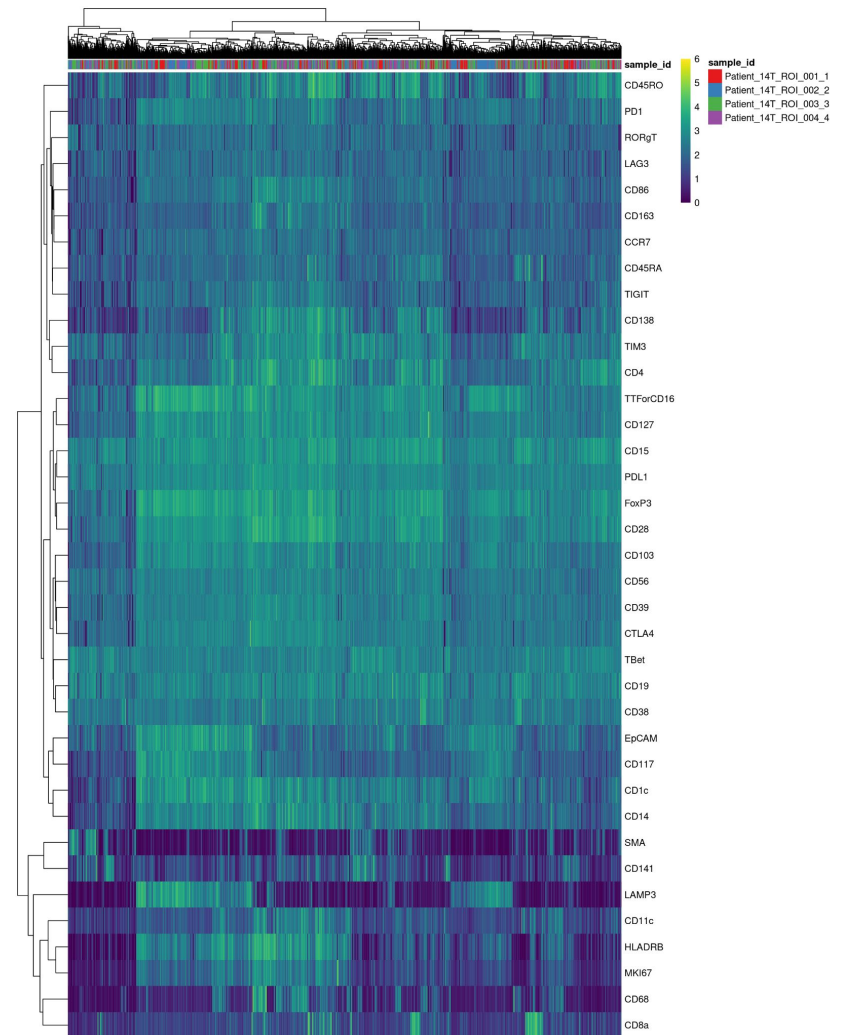
aggregateAcrossCells()
function: computes
average intensity across all
cells per ROI



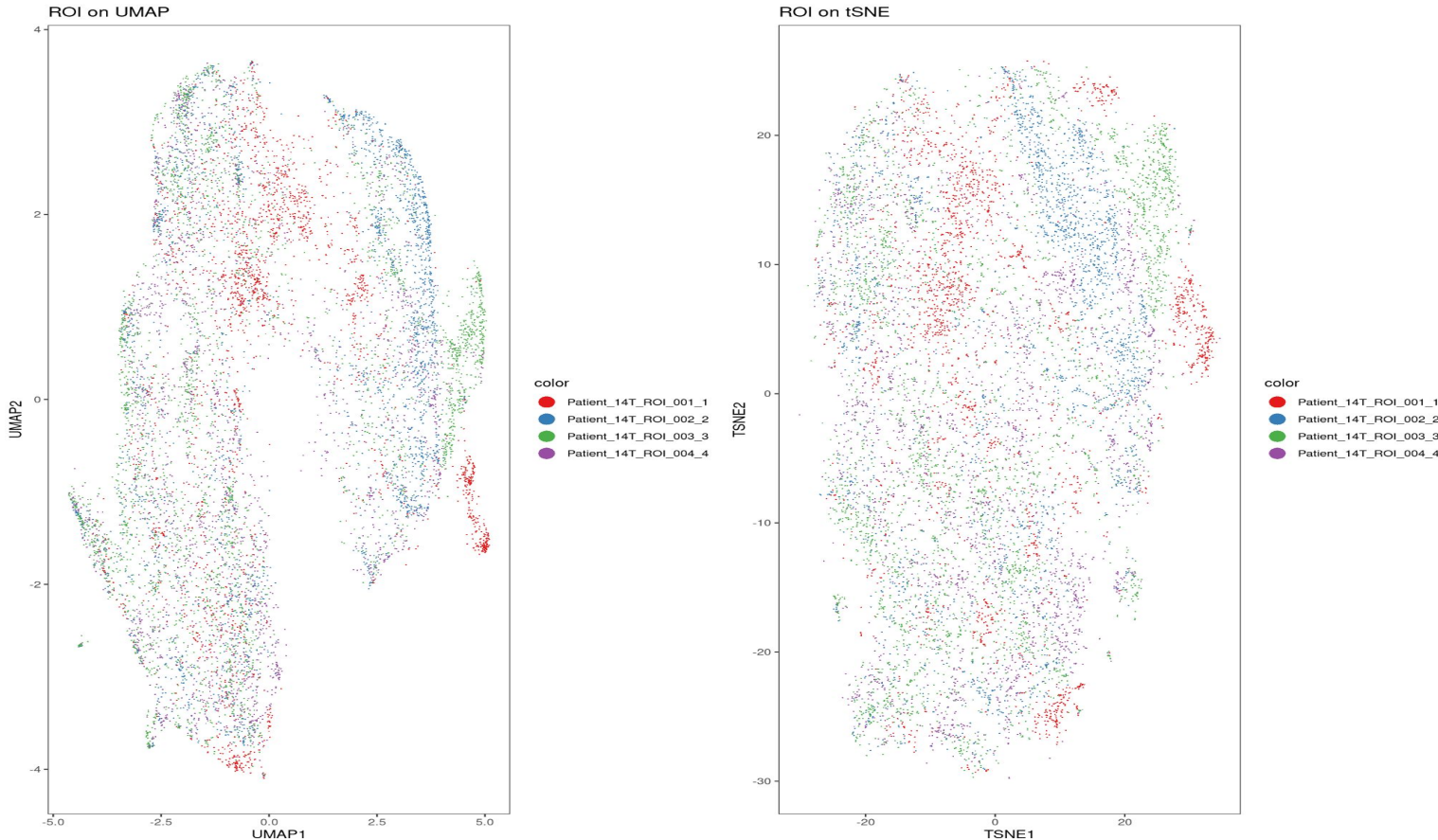
Cell Segmentation - Patient 14 (Necrosis, thus high malignancy)



Patient 14: Heatmap of Single Cell Expression from IMC



Patient 14: ROI UMAP



Questions + Next Steps

Next Steps:

- Distribution after 0-100 (pre-steinbock) → threshold determining (median)
- Run steinbock on all patients
- Run QC metric on all patients (batch effects → observe if correction is needed)
- Cell phenotyping

Questions:

- The 0-100 scaling
- Spillover → anything we can do to address that downstream